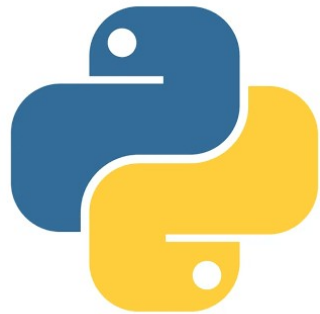


Lecture 5

Intro to Python & Pandas



Announcements

- ❖ Lab 1 today
 - First time getting your hands on a dataset!

Last Time

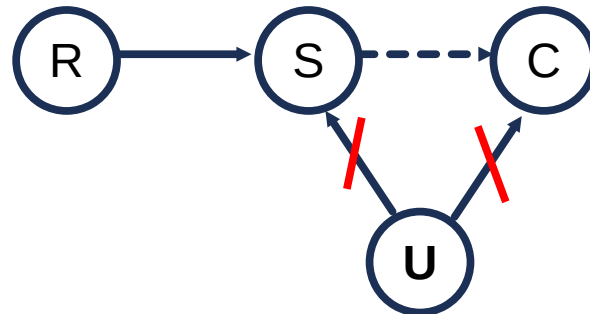
- ❖ Case study: London 1854
- ❖ Confounding
- ❖ Randomization

Today

- ❖ Finish randomization and natural experiments
- ❖ Python from scratch
 - Assigning variables
 - Reading tabular data using Pandas
 - Filtering rows and columns
 - Sorting data

Randomization

- ❖ For a binary treatment (meaning two possible values):
 - **Treatment group**: units receiving treatment
 - **Control group**: units not receiving treatment
- ❖ Idea: randomly assign units to treatment or control
 - Why does this solve our issues with confounding?
 - No other variable U jointly causes both treatment and outcome



Types of Studies

- ❖ **Experimental:** researchers randomize treatment and measure outcome
 - Example: randomized control trial (RCT) for a new cancer drug
 - Not suitable for all domains and may run into practical difficulties

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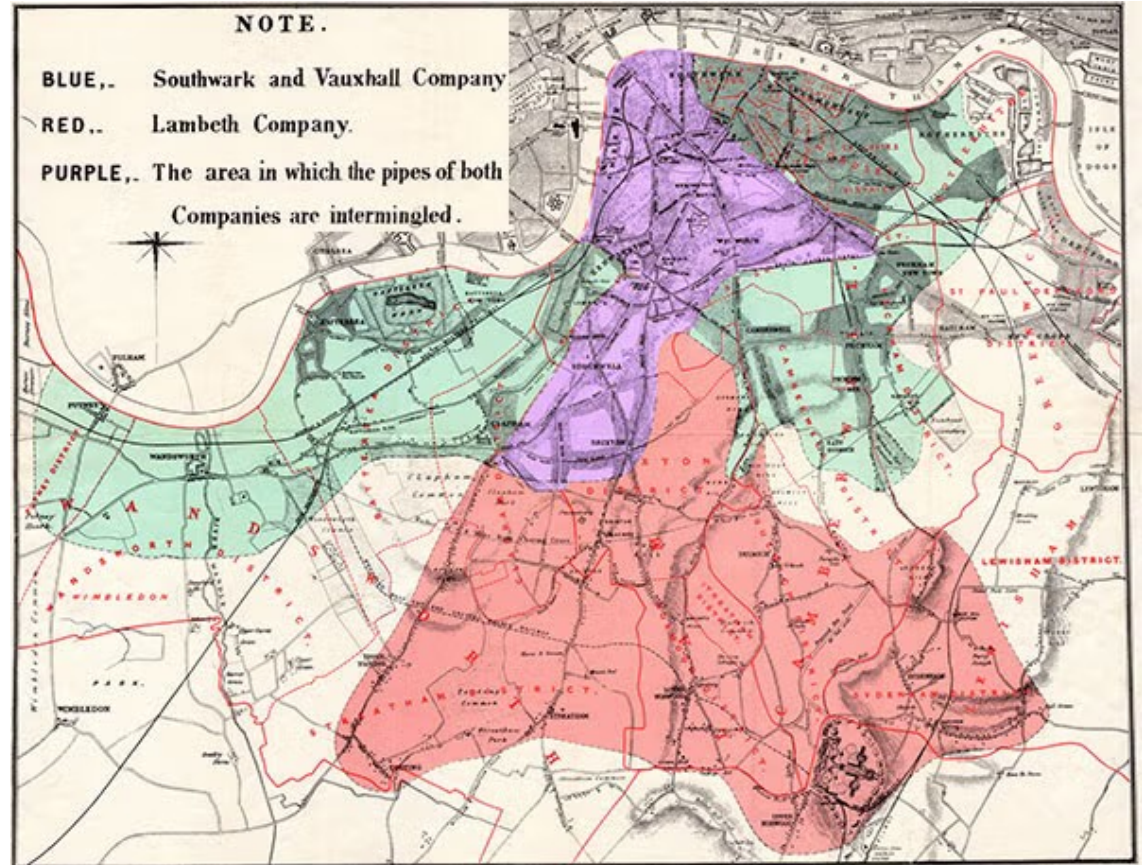
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 - Unless we have more info, restricted to measuring correlations
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 - Example: randomized control trial (RCT) for a new cancer drug
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- ❖ **Observational:** measure treatment and outcome but no intervention
 - Unless we have more info, restricted to measuring correlations
 - Can still be useful!
- ❖ **Natural Experiment:** no intervention but treatment is random-ish
 - Idea: may be something that causes treatment but does not otherwise affect outcome
 - Under some assumptions, we can infer causality

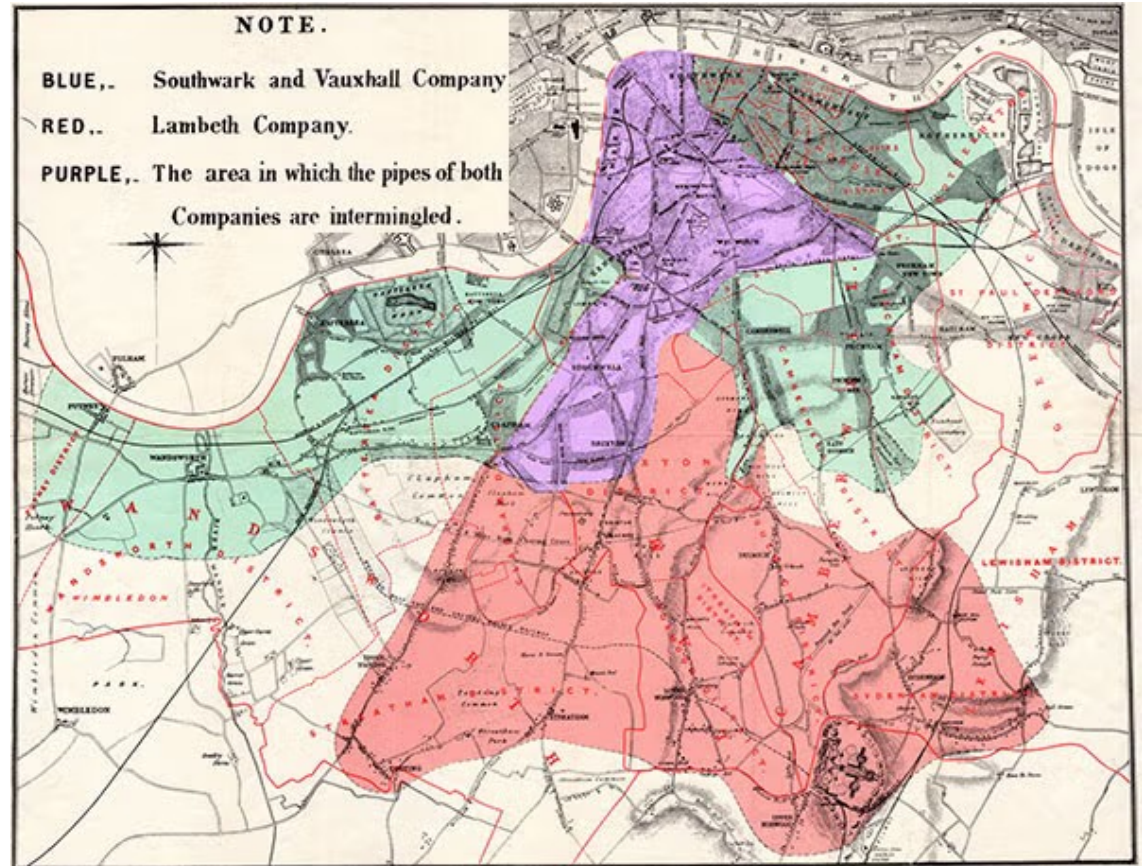
Followup Study

- ❖ In the overlap region, water company was more or less random
- ❖ Lambeth houses and S&V houses otherwise looked alike

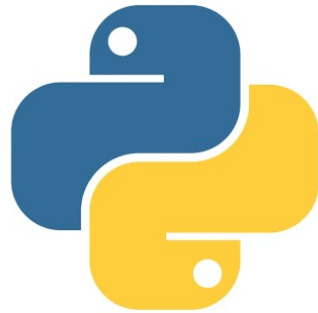


Followup Study

- ❖ In the overlap region, water company was more or less random
- ❖ Lambeth houses and S&V houses otherwise looked alike
- ❖ If the randomization is good enough, we can conclude causality



Python from Scratch



Moving to Jupyter

- ❖ You can follow along in [this notebook](#)

Next Time

- ❖ Read DS Ch. 2
- ❖ Continue working on Python and Pandas